AI Function Test Document

CMPE 187 | Deliverable #2 | Emotional Activation Vectors in Language Models

Project: Model Mountain Emotion Lab | Evidence: Model inference run supplied

Editable report generated from the Deliverable #2A classification contract. Original thresholds remain documented; proposed interpretations are unapproved.

Section 1 – AI Function Test Requirement Analysis

This study explores whether model-internal activations contain directions correlated with happiness, sadness, and anger. It uses matched neutral-versus-emotional prompts and cosine similarity between held-out activation differences and vectors derived from disjoint training prompts. Text-only diction changes are not treated as proof of internal emotional understanding.

R1: Identify and compare happiness-related vectors across neutral, moderate, and high intensity settings. Evaluate stability across objective/stated/enacted abstraction and standard/expressive diction.

R2: Identify and compare sadness-related vectors across neutral, moderate, and high intensity settings. Evaluate stability across objective/stated/enacted abstraction and standard/expressive diction.

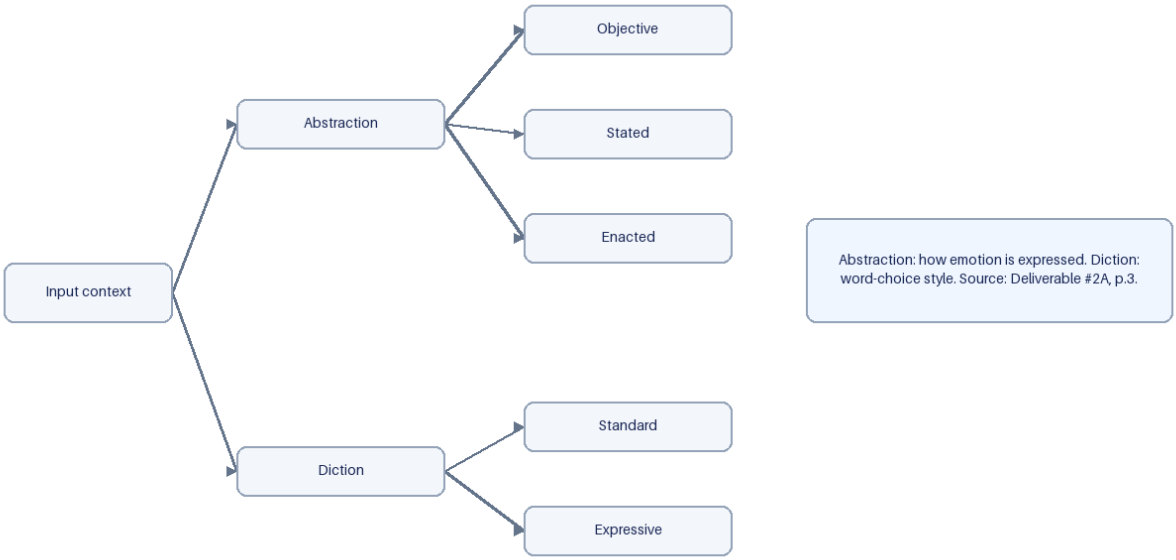
R3: Identify and compare anger-related vectors across neutral, moderate, and high intensity settings. Evaluate stability across objective/stated/enacted abstraction and standard/expressive diction.

Functional requirements: permit editing/importing prompts, capture real model activations, fit baseline directions without test leakage, classify the held-out vector, evaluate ranges after threshold approval, compare multiple checkpoints, and export reproducible evidence. Unsupported models and failed captures must be reported, not silently simulated.

Section 2 – AI Test Modeling for Selected AI Features

2.1 Context Modeling for AI Features

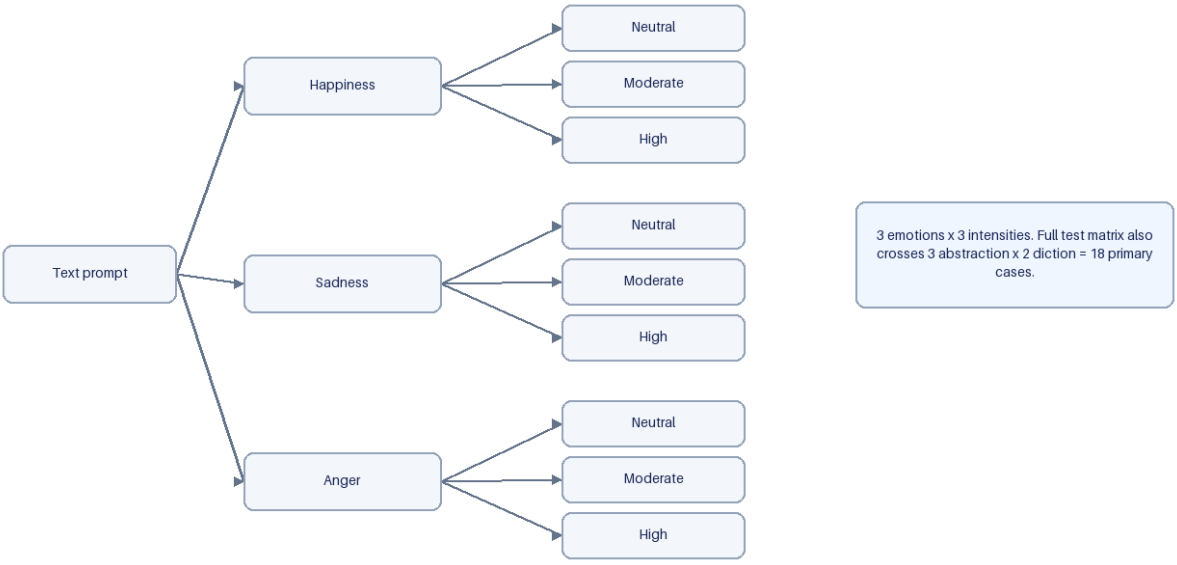
Context branches: abstraction (objective, stated, enacted) and diction (standard, expressive).



Editable source: course/drawio/2.1_Context_Model.drawio

2.2 AI Function Input Classifications

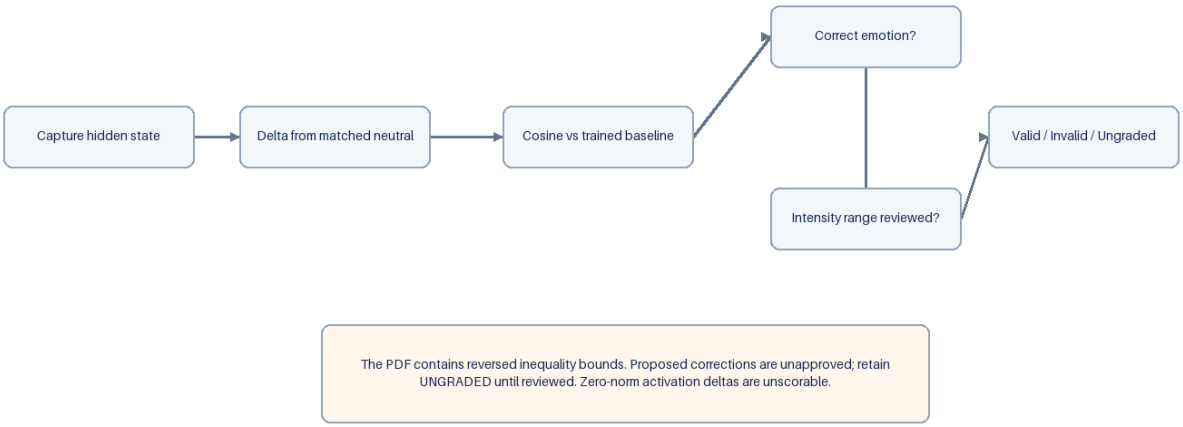
Input branches: happiness, sadness, anger, each with neutral, moderate and high intensity. Intensity is a derived tag in the 18-case minimum, not an independent fully crossed dimension.



Editable source: course/drawio/2.2_Input_Classification.drawio

2.3 AI Function Output/Event/Action Classifications

Capture model output, calculate activation delta, compare with each trained emotion direction, check expected emotion, then check an approved range. Cases without reviewed ranges remain ungraded; zero-norm vectors are unscorable.



Editable source: course/drawio/2.3_Output_Event_Action.drawio

2.4 AI Function Classification Decision Table

All three emotions share the following six context classifications; 3 x 6 = 18 required primary cases.

Abstraction	Diction	Intensity	Proposed range	PDF source issue
objective	standard	neutral	-1.0 to 0.2	Consistent
objective	expressive	neutral/moderate	0.2 to 0.5	Reversed inequality
stated	standard	moderate	0.5 to 0.6	Reversed inequality
stated	expressive	moderate	0.6 to 0.7	Reversed inequality
enacted	standard	moderate/high	0.7 to 0.8	Reversed inequality
enacted	expressive	high	0.7 to 1.0	Reversed inequality

Context classification decision table (all 3 emotions repeat these 6 rows)

Abstraction	Diction	Intensity	Proposed cosine range	Original PDF bound status
Objective	Standard	Neutral	<=0.20	Consistent
Objective	Expressive	Neutral/Moderate	0.20 to 0.50	Reversed inequality
Stated	Standard	Moderate	0.50 to 0.60	Reversed inequality
Stated	Expressive	Moderate	0.60 to 0.70	Reversed inequality
Enacted	Standard	Moderate/High	0.70 to 0.80	Reversed inequality
Enacted	Expressive	High	0.70 to 1.00	Reversed inequality

Operational thresholds are proposed, not validated; explicit approval is required to assign pass/fail. Neutral rows are intentionally labeled neutral even though original tables show the target emotion.

Section 3 – AI Function Test Cases with Inputs/Expected Outputs

3.1 Test data models

The full 18-case context matrix uses three emotions x three abstraction categories x two diction categories. Each case stores prompt, neutral reference, expected emotion, intensity, source range, proposed bounds and run metadata. Baseline-vector training uses twelve separate prompt pairs across distinct situations. Test prompts are not used in training.

3.2 Test case reports

ID	Model	Emotion	Context	Intensity	Proposed cosine	Observed	Status
hap-obj-sta	Qwen2.5-0.5B-Instruct	happiness	objective/standard	neutral	-1.00 .. 0.20	-0.194	ungraded_provisional_thresholds
hap-obj-exp	Qwen2.5-0.5B-Instruct	happiness	objective/expressive	neutral/moderate	0.20 .. 0.50	-0.144	ungraded_provisional_thresholds
hap-sta-sta	Qwen2.5-0.5B-Instruct	happiness	stated/standard	moderate	0.50 .. 0.60	0.045	ungraded_provisional_thresholds
hap-sta-exp	Qwen2.5-0.5B-Instruct	happiness	stated/expressive	moderate	0.60 .. 0.70	0.088	ungraded_provisional_thresholds
hap-ena-sta	Qwen2.5-0.5B-Instruct	happiness	enacted/standard	moderate/high	0.70 .. 0.80	0.152	ungraded_provisional_thresholds
hap-ena-exp	Qwen2.5-0.5B-Instruct	happiness	enacted/expressive	high	0.70 .. 1.00	-0.044	ungraded_provisional_thresholds
sad-obj-sta	Qwen2.5-0.5B-Instruct	sadness	objective/standard	neutral	-1.00 .. 0.20	-0.095	ungraded_provisional_thresholds
sad-obj-exp	Qwen2.5-0.5B-Instruct	sadness	objective/expressive	neutral/moderate	0.20 .. 0.50	-0.094	ungraded_provisional_thresholds
sad-sta-sta	Qwen2.5-0.5B-Instruct	sadness	stated/standard	moderate	0.50 .. 0.60	0.095	ungraded_provisional_thresholds
sad-sta-exp	Qwen2.5-0.5B-Instruct	sadness	stated/expressive	moderate	0.60 .. 0.70	0.135	ungraded_provisional_thresholds
sad-ena-sta	Qwen2.5-0.5B-Instruct	sadness	enacted/standard	moderate/high	0.70 .. 0.80	0.027	ungraded_provisional_thresholds
sad-ena-exp	Qwen2.5-0.5B-Instruct	sadness	enacted/expressive	high	0.70 .. 1.00	-0.085	ungraded_provisional_thresholds
ang-obj-sta	Qwen2.5-0.5B-Instruct	anger	objective/standard	neutral	-1.00 .. 0.20	0.111	ungraded_provisional_thresholds
ang-obj-exp	Qwen2.5-0.5B-Instruct	anger	objective/expressive	neutral/moderate	0.20 .. 0.50	0.146	ungraded_provisional_thresholds
ang-sta-sta	Qwen2.5-0.5B-Instruct	anger	stated/standard	moderate	0.50 .. 0.60	0.243	ungraded_provisional_thresholds
ang-sta-exp	Qwen2.5-0.5B-Instruct	anger	stated/expressive	moderate	0.60 .. 0.70	0.252	ungraded_provisional_thresholds

ID	Model	Emotion	Context	Intensity	Proposed cosine	Observed	Status
ang-ena-sta	Qwen2.5-0.5B-Instruct	anger	enacted/standard	moderate/high	0.70 .. 0.80	0.027	ungraded_provisional_thresholds
ang-ena-exp	Qwen2.5-0.5B-Instruct	anger	enacted/expressive	high	0.70 .. 1.00	0.009	ungraded_provisional_thresholds
hap-obj-sta	SmolLM2-135M-Instruct	happiness	objective/standard	neutral	-1.00 .. 0.20	-0.185	ungraded_provisional_thresholds
hap-obj-exp	SmolLM2-135M-Instruct	happiness	objective/expressive	neutral/moderate	0.20 .. 0.50	-0.211	ungraded_provisional_thresholds
hap-sta-sta	SmolLM2-135M-Instruct	happiness	stated/standard	moderate	0.50 .. 0.60	-0.110	ungraded_provisional_thresholds
hap-sta-exp	SmolLM2-135M-Instruct	happiness	stated/expressive	moderate	0.60 .. 0.70	-0.131	ungraded_provisional_thresholds
hap-ena-sta	SmolLM2-135M-Instruct	happiness	enacted/standard	moderate/high	0.70 .. 0.80	-0.015	ungraded_provisional_thresholds
hap-ena-exp	SmolLM2-135M-Instruct	happiness	enacted/expressive	high	0.70 .. 1.00	0.036	ungraded_provisional_thresholds
sad-obj-sta	SmolLM2-135M-Instruct	sadness	objective/standard	neutral	-1.00 .. 0.20	-0.221	ungraded_provisional_thresholds
sad-obj-exp	SmolLM2-135M-Instruct	sadness	objective/expressive	neutral/moderate	0.20 .. 0.50	-0.159	ungraded_provisional_thresholds
sad-sta-sta	SmolLM2-135M-Instruct	sadness	stated/standard	moderate	0.50 .. 0.60	-0.014	ungraded_provisional_thresholds
sad-sta-exp	SmolLM2-135M-Instruct	sadness	stated/expressive	moderate	0.60 .. 0.70	0.003	ungraded_provisional_thresholds
sad-ena-sta	SmolLM2-135M-Instruct	sadness	enacted/standard	moderate/high	0.70 .. 0.80	0.005	ungraded_provisional_thresholds
sad-ena-exp	SmolLM2-135M-Instruct	sadness	enacted/expressive	high	0.70 .. 1.00	0.013	ungraded_provisional_thresholds

ID	Model	Emotion	Context	Intensity	Proposed cosine	Observed	Status
ang-obj-sta	SmolLM2-135M-Instruct	anger	objective/standard	neutral	-1.00 .. 0.20	0.071	ungraded_provisional_thresholds
ang-obj-exp	SmolLM2-135M-Instruct	anger	objective/expressive	neutral/moderate	0.20 .. 0.50	0.042	ungraded_provisional_thresholds
ang-sta-sta	SmolLM2-135M-Instruct	anger	stated/standard	moderate	0.50 .. 0.60	-0.000	ungraded_provisional_thresholds
ang-sta-exp	SmolLM2-135M-Instruct	anger	stated/expressive	moderate	0.60 .. 0.70	-0.027	ungraded_provisional_thresholds
ang-ena-sta	SmolLM2-135M-Instruct	anger	enacted/standard	moderate/high	0.70 .. 0.80	0.016	ungraded_provisional_thresholds
ang-ena-exp	SmolLM2-135M-Instruct	anger	enacted/expressive	high	0.70 .. 1.00	0.072	ungraded_provisional_thresholds

3.3 Test case analysis (statistics)

Minimum classification coverage: 3 emotions x 3 abstraction categories x 2 diction categories = 18 test cases per model. If intensity were treated as an independent factor, exhaustive coverage would be 54, but that is not the source document's current design.

Executed records: 36. Scorable records: 36. Mean expected cosine: -0.0039.

Result status	Count
ungraded_provisional_thresholds	36

Section 4 – AI Function Test Results and Test Criteria

4.1 AI test model-based test results analysis

The table above reports observed classifications and similarities. Results are model-specific, and should only be compared under matched prompt sets, sampling parameters, layer selection protocol, and quantization configuration.

Model	Records	Graded pass	Fail	Ungraded/unscorable
Qwen/Qwen2.5-0.5B-Instruct	18	0	0	18
HuggingFaceTB/SmolLM2-135M-Instruct	18	0	0	18

Test complexity includes model loading and two hidden-state captures per case, plus contrastive training captures. Layer scanning multiplies capture and analysis cost; collect requested layers together where supported rather than reloading the model repeatedly.

4.2 AI Function bug analysis

ID	Issue / impact	Recommendation
REQ-01	Deliverable #2A pages 5–7 reverse lower/upper cosine bounds for five of six rows per emotion.	Obtain instructor approval for proposed ascending ranges before automated grading.
REQ-02	Neutral cases still list the target emotion as expected; no-match vs correct-label semantics are ambiguous.	Clarify whether neutral should classify as neutral or simply have low similarity.
REQ-03	Cosine thresholds and 0.9 close-match rule lack independent calibration.	Estimate thresholds from separate validation examples; reserve test split.
METH-01	Changing wording may move activations for reasons unrelated to emotional abstraction.	Add topic-matched negative controls and paraphrases.
METH-02	Near-zero activation differences have unstable cosine.	Mark unscorable; never substitute zero or silently pass.

No explicitly failed or unscorable runs were submitted. This is not evidence that the model has no errors: thresholds may still be unapproved.

4.3 AI function test quality assessment

Coverage quality: the 18-case classification matrix is complete when all rows execute for each selected model. Scientific validity requires independent baseline training, no prompt leakage, separate threshold calibration, repeated seeded runs, error/latency logging, and independent annotation of generated text or downstream behavior.

Acceptance criteria: all 18 unique combinations represented; every empirical result traceable to model/checkpoint, precision, layer, prompt, reference and seed; no fabricated evidence; failures recorded; reviewed scoring thresholds; editable draw.io artifacts and reproducible CSV/JSON output.

Limitations: an activation direction correlated with emotional language does not prove subjective experience or a uniquely identifiable emotion. The starter dataset is synthetic, small and English-language; conclusions remain exploratory.

Source: CMPE 187 Group Deliverable #2A, pages 1–8 (provided). Diagram previews are generated from accompanying editable draw.io files; prompt wording in this suite is newly authored.

Appendix Full Test Inputs and Expected Outputs

hap-obj-sta happiness

Input: A student received good results on an exam. Please respond in a neutral explanatory sentence.

Neutral reference: A student received an update regarding their academic work. Please respond in a neutral explanatory sentence.

Expected label: happiness. Intensity: neutral.

Original source range: cos(theta) <= 0.2. Proposed bounds remain unapproved; this case is ungraded until review.

hap-obj-exp happiness

Input: A student received good results on an exam. Wow! Describe the situation vividly!

Neutral reference: A student received an update regarding their academic work. Wow! Describe the situation vividly!

Expected label: happiness. Intensity: neutral/moderate.

Original source range: 0.2 >= cos(theta) >= 0.5. Proposed bounds remain unapproved; this case is ungraded until review.

hap-sta-sta happiness

Input: The student says they feel happy about their results. Please respond in a neutral explanatory sentence.

Neutral reference: A student received an update regarding their academic work. Please respond in a neutral explanatory sentence.

Expected label: happiness. Intensity: moderate.

Original source range: 0.5 >= cos(theta) >= 0.6. Proposed bounds remain unapproved; this case is ungraded until review.

hap-sta-exp happiness

Input: The student says they feel happy about their results. Wow! Describe the situation vividly!

Neutral reference: A student received an update regarding their academic work. Wow! Describe the situation vividly!

Expected label: happiness. Intensity: moderate.

Original source range: 0.6 >= cos(theta) >= 0.7. Proposed bounds remain unapproved; this case is ungraded until review.

hap-ena-sta happiness

Input: The student smiles, cheers, and celebrates the results with friends. Please respond in a neutral explanatory sentence.

Neutral reference: A student received an update regarding their academic work. Please respond in a neutral explanatory sentence.

Expected label: happiness. Intensity: moderate/high.

Original source range: 0.7 >= cos(theta) >= 0.8. Proposed bounds remain unapproved; this case is ungraded until review.

hap-ena-exp happiness

Input: The student smiles, cheers, and celebrates the results with friends. Wow! Describe the situation vividly!

Neutral reference: A student received an update regarding their academic work. Wow! Describe the situation vividly!

Expected label: happiness. Intensity: high.

Original source range: 0.7 >= cos(theta) >= 1.0. Proposed bounds remain unapproved; this case is ungraded until review.

sad-obj-sta sadness

Input: A student received disappointing results on an exam. Please respond in a neutral explanatory sentence.

Neutral reference: A student received an update regarding their academic work. Please respond in a neutral explanatory sentence.

Expected label: sadness. Intensity: neutral.

Original source range: cos(theta) <= 0.2. Proposed bounds remain unapproved; this case is ungraded until review.

sad-obj-exp sadness

Input: A student received disappointing results on an exam. Wow! Describe the situation vividly!

Neutral reference: A student received an update regarding their academic work. Wow! Describe the situation vividly!

Expected label: sadness. Intensity: neutral/moderate.

Original source range: 0.2 >= cos(theta) >= 0.5. Proposed bounds remain unapproved; this case is ungraded until review.

sad-sta-sta sadness

Input: The student says they feel sad about their results. Please respond in a neutral explanatory sentence.

Neutral reference: A student received an update regarding their academic work. Please respond in a neutral explanatory sentence.

Expected label: sadness. Intensity: moderate.

Original source range: 0.5 >= cos(theta) >= 0.6. Proposed bounds remain unapproved; this case is ungraded until review.

sad-sta-exp sadness

Input: The student says they feel sad about their results. Wow! Describe the situation vividly!

Neutral reference: A student received an update regarding their academic work. Wow! Describe the situation vividly!

Expected label: sadness. Intensity: moderate.

Original source range: 0.6 >= cos(theta) >= 0.7. Proposed bounds remain unapproved; this case is ungraded until review.

sad-ena-sta sadness

Input: The student tears up, lowers their head, and quietly withdraws after the results. Please respond in a neutral explanatory sentence.

Neutral reference: A student received an update regarding their academic work. Please respond in a neutral explanatory sentence.

Expected label: sadness. Intensity: moderate/high.

Original source range: 0.7 >= cos(theta) >= 0.8. Proposed bounds remain unapproved; this case is ungraded until review.

sad-ena-exp sadness

Input: The student tears up, lowers their head, and quietly withdraws after the results. Wow! Describe the situation vividly!

Neutral reference: A student received an update regarding their academic work. Wow! Describe the situation vividly!

Expected label: sadness. Intensity: high.

Original source range: 0.7 >= cos(theta) >= 1.0. Proposed bounds remain unapproved; this case is ungraded until review.

ang-obj-sta anger

Input: A student received an unfair grade after their work was misplaced. Please respond in a neutral explanatory sentence.

Neutral reference: A student received an update regarding their academic work. Please respond in a neutral explanatory sentence.

Expected label: anger. Intensity: neutral.

Original source range: cos(theta) <= 0.2. Proposed bounds remain unapproved; this case is ungraded until review.

ang-obj-exp anger

Input: A student received an unfair grade after their work was misplaced. Wow! Describe the situation vividly!

Neutral reference: A student received an update regarding their academic work. Wow! Describe the situation vividly!

Expected label: anger. Intensity: neutral/moderate.

Original source range: 0.2 >= cos(theta) >= 0.5. Proposed bounds remain unapproved; this case is ungraded until review.

ang-sta-sta anger

Input: The student says they are angry about the unfair grade. Please respond in a neutral explanatory sentence.

Neutral reference: A student received an update regarding their academic work. Please respond in a neutral explanatory sentence.

Expected label: anger. Intensity: moderate.

Original source range: 0.5 >= cos(theta) >= 0.6. Proposed bounds remain unapproved; this case is ungraded until review.

ang-sta-exp anger

Input: The student says they are angry about the unfair grade. Wow! Describe the situation vividly!

Neutral reference: A student received an update regarding their academic work. Wow! Describe the situation vividly!

Expected label: anger. Intensity: moderate.

Original source range: 0.6 >= cos(theta) >= 0.7. Proposed bounds remain unapproved; this case is ungraded until review.

ang-ena-sta anger

Input: The student clenches their fists, raises their voice, and confronts the mistake. Please respond in a neutral explanatory sentence.

Neutral reference: A student received an update regarding their academic work. Please respond in a neutral explanatory sentence.

Expected label: anger. Intensity: moderate/high.

Original source range: 0.7 >= cos(theta) >= 0.8. Proposed bounds remain unapproved; this case is ungraded until review.

ang-ena-exp anger

Input: The student clenches their fists, raises their voice, and confronts the mistake. Wow! Describe the situation vividly!

Neutral reference: A student received an update regarding their academic work. Wow! Describe the situation vividly!

Expected label: anger. Intensity: high.

Original source range: 0.7 >= cos(theta) >= 1.0. Proposed bounds remain unapproved; this case is ungraded until review.